

Blockhouse Task - Conceptual Answers

Task 1 — Order-Flow Imbalance

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1. Why measure OFI at *multiple* depth levels?

Relying on the best bid and ask captures only the 'visible tip' of liquidity, obscuring latent supply and demand hidden a few ticks deeper, where large orders often sit. Including those additional layers reveals genuine supply–demand pressure. Ignoring those layers understates the true imbalance. Empirically, the paper confirms that extending OFI from one to ten levels lifts the explanatory power of the sample by roughly 15 percentage points, underscoring the incremental information that deeper book data contribute. In conclusion, multi-level OFI captures hidden and anticipatory order flow that moves prices before it surfaces at the quote.

2. Why employ Lasso regression rather than OLS for cross-impact?

Estimating cross-impact among N stocks involves $N(N-1)$ potential influence coefficients; even with one-minute data, an ordinary least squares regression would be badly overparameterized and highly unstable. The order-flow vectors of related names are also strongly collinear, inflating OLS variance and yielding coefficients that are neither robust nor interpretable. By imposing a penalty ℓ_1 , Lasso shrinks most coefficients to zero, automatically selecting the handful of 'leader' stocks that genuinely drive short-run price moves in others while eliminating noise. The result is a sparse cross-impact matrix that is easier to monitor in real time and less prone to overfitting. In conclusion, Lasso tames the high-dimensional, collinear cross-impact problem and produces interpretable and robust influence maps.

3. Why does OFI out-predict raw trade volume at short horizons?

Volume is agnostic to buy vs. sell pressure; a million shares traded tells us nothing about *who* was more aggressive. OFI is *signed*: additions/cancellations on the bid *minus* those on the ask directly encode net buying pressure. Also, large but offsetting buy/sell trades inflate volume, yet leave price unchanged. OFI nets those flows out, raising signal-to-noise when forecasting the next few seconds or minutes. In conclusion, by embedding both direction and timing, OFI provides a cleaner, earlier view of supply–demand imbalances than raw volume, yielding stronger short-run return forecasts.